

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1596

COURSE OUTCOME COVERED

CO3: Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I **G Althaf (192372004)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

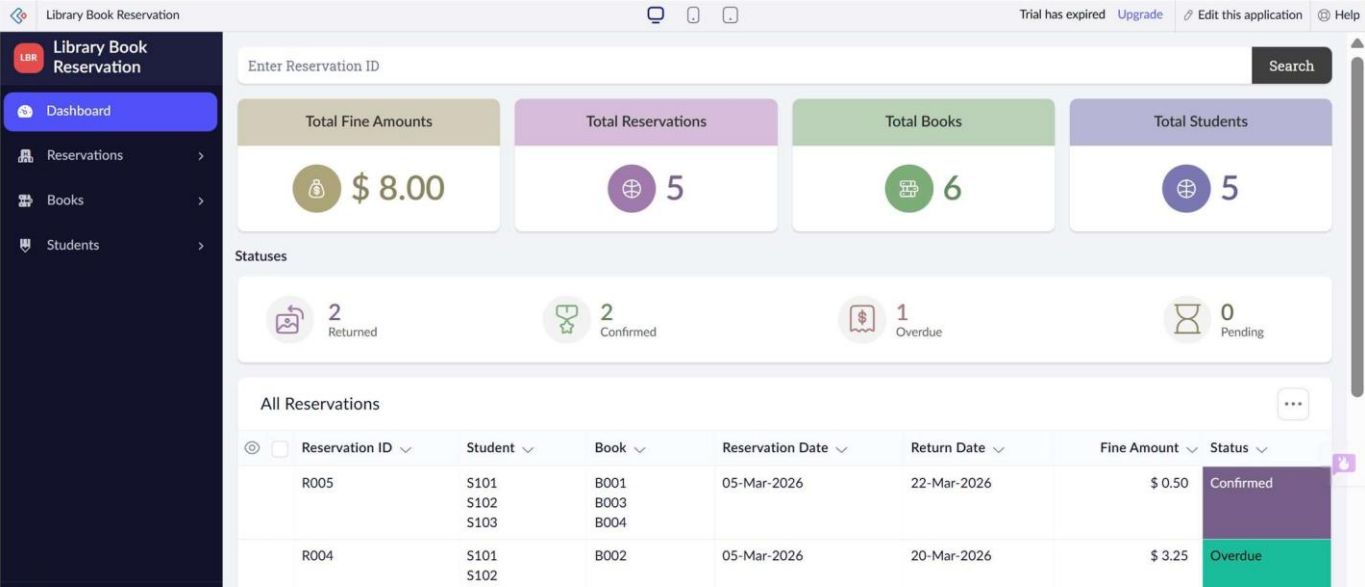
Total Marks Awarded:

Assessment Tasks-1

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Screenshots:

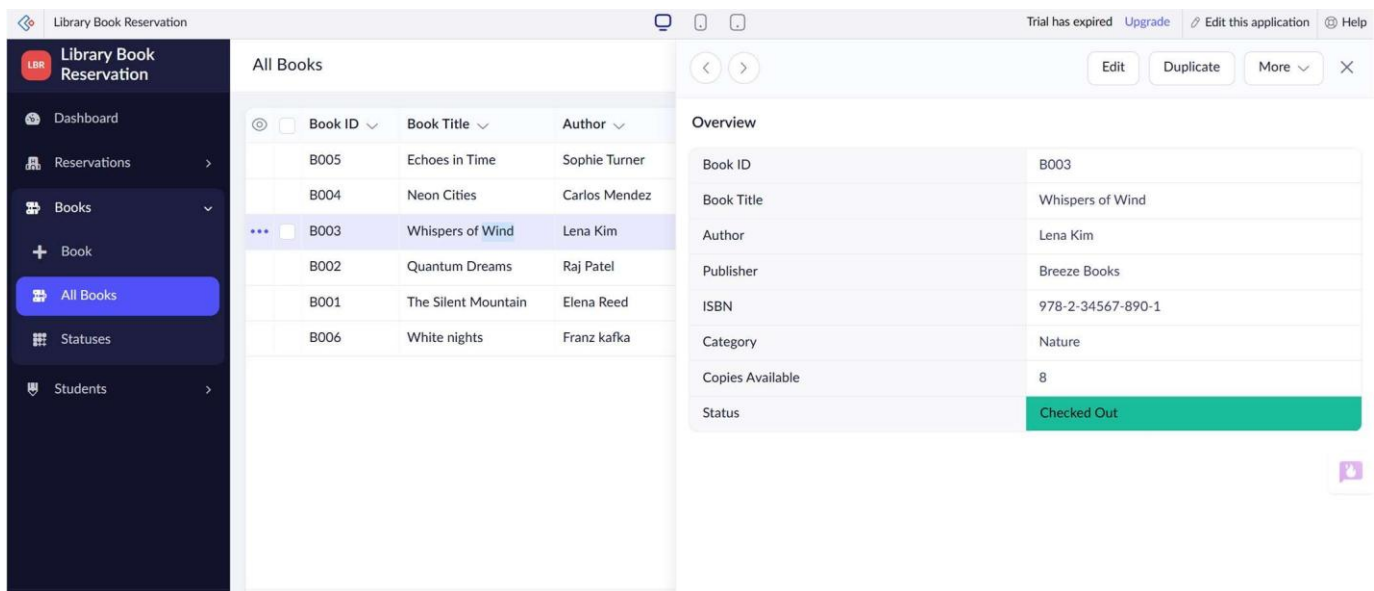


Library Book Reservation

All Books

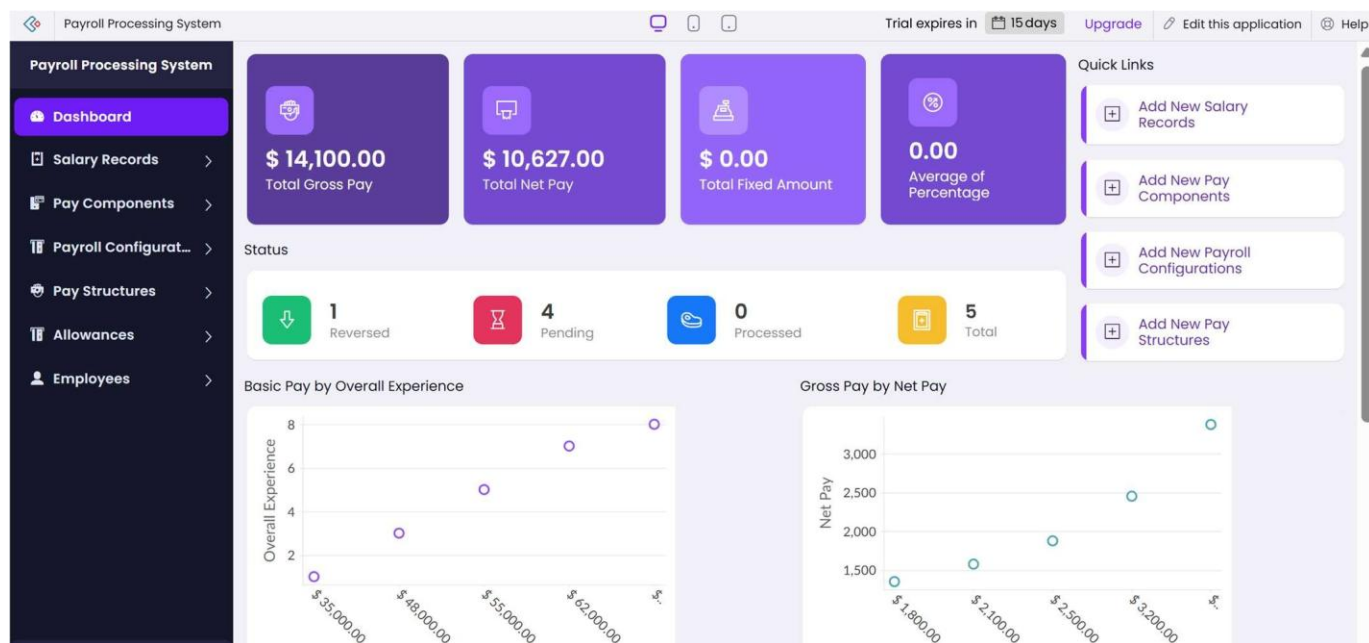
Q + ...

Book ID	Book Title	Author	Publisher	ISBN	Category	Copies Available	Status
B005	Echoes in Time	Sophie Turner	Chrono Publishing	978-4-56789-012-3	History	7	Unavailable
B004	Neon Cities	Carlos Mendez	Urban Edge	978-3-45678-901-2	Urban Fiction	2	Reserved
B003	Whispers of Wind	Lena Kim	Breeze Books	978-2-34567-890-1	Nature	8	Checked Out
B002	Quantum Dreams	Raj Patel	Nova Science	978-0-98765-432-1	Science	3	Unavailable
B001	The Silent Mountain	Elena Reed	Sunrise Press	978-1-23456-789-0	Fiction	5	Available
B006	White nights	Franz kafka	William publications	124824141	Fiction	3	Available



Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Screenshots:



Payroll Processing System

Trial expires in 15 days Upgrade Edit this application Help

Payroll Processing System

Payroll Configurations

Dashboard

Salary Records >

Pay Components >

Payroll Configurati... >

+ Payroll Configurati...

Payroll Configurati...

Pay Structures >

Allowances >

Employees >

Structure

Component ~Select-

Calculation Order #####

☐ Is Required

Description

Submit Reset

Payroll Processing System

Trial expires in 15 days Upgrade Edit this application Help

Payroll Processing System

Employees

Dashboard

Salary Records >

Pay Components >

Payroll Configurati... >

Pay Structures >

Allowances >

Employees >

+ Employees

Employees Report

Board

Name

Experience in Present Org #####

Overall Experience #####

Basic Pay #####.###

Hire Date dd-MMM-yyyy

Status ~Select- v

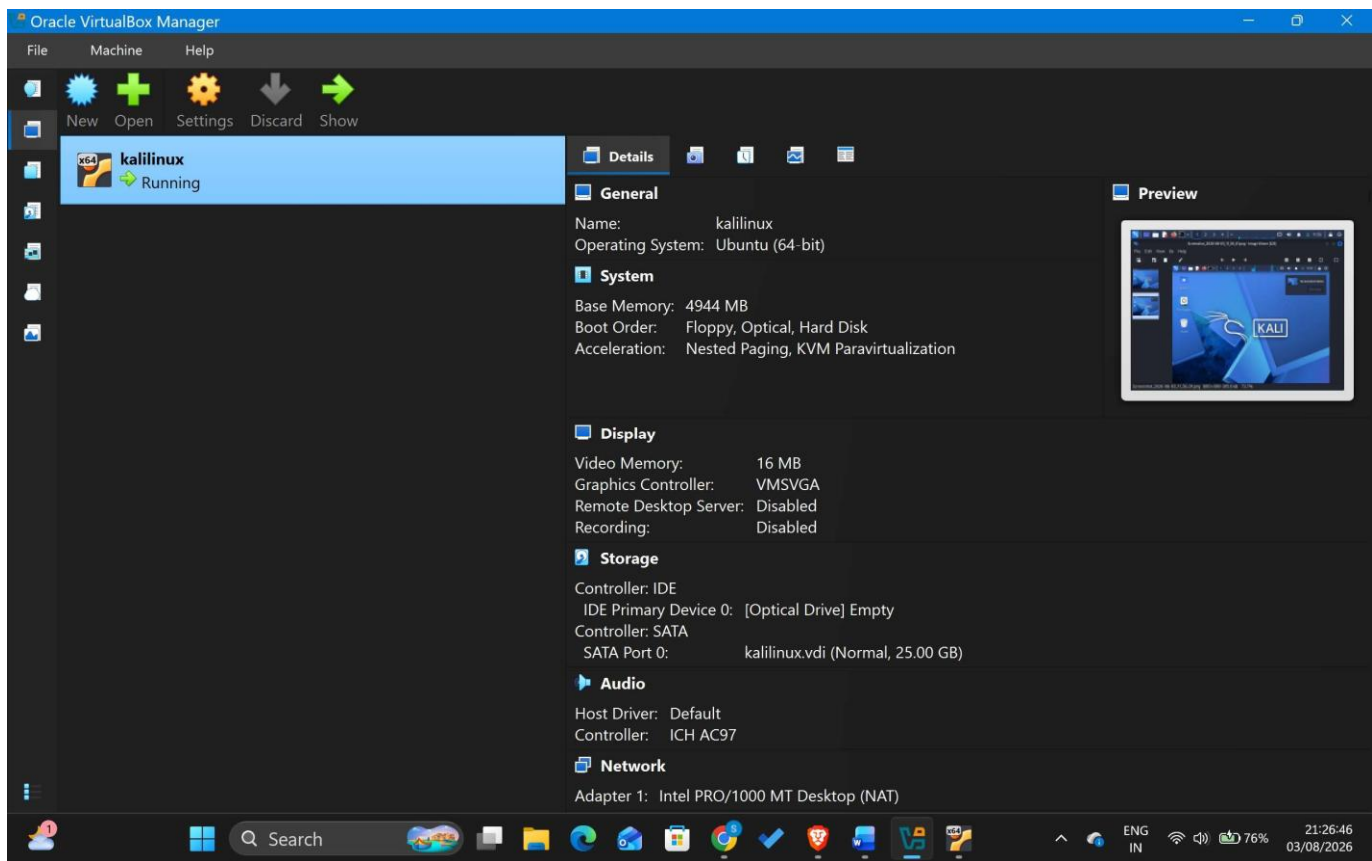
Department

User Lookup

Submit Reset

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Screenshots:





Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Screenshots:

